

SPIRIT OF ST LOUIS, MO, USA

WMO: 724345

Lat: 38.657N

Lon: 90.656W

Elev: 462

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	11.3	-4.2	4.5	9.2	0.4	5.7	13.8	24.5	34.3	21.5	34.2	6.4	290	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	95.4	77.3	92.9	76.5	90.5	75.4	79.9	90.7	78.4	89.0	77.3	87.3	7.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	143.0	85.6	75.4	135.2	84.5	74.2	129.7	83.0	43.7	90.8	42.3	89.1	40.9	87.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	18.3	16.3	DB	-2.6	99.9	6.0	3.4	-6.9	102.3	-10.4	104.3	-13.7	106.2	-18.1	108.7
			WB	-3.3	82.2	5.8	1.5	-7.5	83.3	-10.8	84.2	-14.1	85.1	-18.3	86.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	31.5	35.9	45.9	56.4	66.0	74.7	78.5	77.0	68.9	57.4	45.5	35.5
	DBStd	18.66	11.84	11.08	10.94	9.41	8.08	6.08	5.60	5.49	7.50	9.12	9.77	10.32
	HDD50	1941	584	413	211	40	2	0	0	0	0	30	200	461
	HDD65	4651	1040	815	597	287	90	6	0	1	45	269	587	914
	CDD50	4206	9	18	84	232	498	742	882	837	567	261	64	12
	CDD65	1445	0	0	6	29	122	298	418	373	162	35	2	0
	CDH74	14911	0	4	72	350	1116	2980	4610	3758	1628	368	25	0
	CDH80	6199	0	0	8	85	339	1231	2164	1666	612	92	2	0
(14) Wind	WSAvg	6.4	7.7	7.8	7.9	8.2	6.4	5.5	5.0	4.5	4.5	5.9	6.9	7.0
(15) Precipitation	PrecAvg	43.2	2.6	2.2	3.3	4.7	5.2	4.7	3.8	3.5	3.5	3.2	3.5	2.8
	PrecMax	61.1	8.7	4.6	7.0	11.6	11.9	10.0	7.6	6.2	12.0	13.7	7.7	10.2
	PrecMin	27.1	0.7	0.4	1.4	1.8	1.3	1.1	1.1	1.4	0.6	0.8	0.9	0.7
	PrecStd	7.8	1.8	1.2	1.3	2.2	2.8	2.3	1.8	1.4	2.5	2.5	2.1	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	72.5	80.8	87.4	90.4	95.7	100.7	98.9	93.6	87.4	77.7	69.3
		MCWB	58.6	58.3	62.9	66.9	72.7	76.1	76.4	76.8	74.1	71.2	64.1	61.2
	2%	DB	61.3	66.2	75.5	82.1	87.2	92.6	96.1	94.9	90.4	82.3	71.8	63.2
		MCWB	54.1	55.4	60.6	63.8	71.2	76.2	78.2	77.1	73.9	67.6	59.0	56.5
	5%	DB	55.3	60.9	70.3	78.3	83.8	90.2	93.1	91.6	87.1	78.2	66.8	57.2
		MCWB	47.9	51.9	57.4	62.6	70.0	75.4	77.3	75.9	72.2	65.1	57.7	50.2
	10%	DB	49.2	54.8	64.9	73.5	80.7	87.5	90.4	88.7	83.0	73.6	62.9	52.1
		MCWB	43.2	47.1	54.8	60.5	68.3	73.7	76.3	74.6	69.9	63.2	55.1	46.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	61.6	65.8	69.3	75.9	80.1	82.3	81.2	77.7	73.5	65.8	62.4
		MCDB	66.9	68.3	77.6	81.2	86.0	90.4	92.3	91.8	88.9	83.2	72.9	67.9
	2%	WB	55.3	57.9	62.3	66.7	73.7	78.1	80.4	79.3	75.9	70.7	62.3	57.4
		MCDB	60.2	64.8	72.0	77.5	83.8	88.9	91.0	90.3	86.7	78.7	68.3	62.3
	5%	WB	49.1	52.7	59.7	64.7	71.7	76.8	79.0	77.9	74.3	67.8	59.5	51.7
		MCDB	55.1	59.5	68.0	74.5	80.7	87.1	89.7	88.1	83.3	74.8	65.7	55.6
	10%	WB	43.1	46.6	56.1	62.4	69.9	75.2	77.8	76.4	72.5	64.8	55.5	46.4
		MCDB	49.0	54.3	64.3	71.5	78.3	84.8	87.7	85.7	80.1	72.1	62.4	50.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.9	21.4	22.6	20.7	20.1	20.2	20.8	23.1	22.8	20.6	17.6
		MCDBR	27.7	29.6	28.6	28.7	23.7	22.9	22.6	23.8	26.0	27.5	27.7	25.6
		MCWBR	19.7	19.9	16.0	14.7	11.1	9.7	8.6	9.0	11.0	14.2	17.8	18.7
	5% WB	MCDBR	24.1	26.5	24.3	23.6	20.5	20.6	20.3	20.7	21.5	21.7	23.6	23.2
		MCWBR	18.9	19.9	15.6	13.9	10.6	9.7	8.9	8.8	9.7	12.9	17.4	19.0
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.345	0.399	0.439	0.449	0.475	0.466	0.408	0.352	0.311	0.303
	taud		2.477	2.428	2.419	2.263	2.211	2.220	2.148	2.178	2.312	2.475	2.531	2.496
	Ebn at Noon		278	288	290	279	268	265	257	256	265	272	274	270
	Edh at Noon		25	30	34	42	45	45	48	45	37	29	24	23
(44) All-Sky Solar Radiation	RadAvg		655	877	1188	1572	1768	2013	1971	1760	1511	1085	737	553
	RadStd		74	88	115	92	155	156	136	100	95	120	75	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+148
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page